Andrey Violante

Backend Developer · Python

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Maricá, RJ, Brazil · open to remote

SUMMARY

Software Engineering student in the 6th semester, focused on backend: Python APIs, relational and document databases, messaging and containers. I built an AI app used in classrooms and an open-source tool for robotics classes with its own compiler. Robotics teacher since 2023. Looking for a backend internship or junior role.

PROJECTS

EST-LAB · Python, reverse engineering, compilers
github.com/AndreyViolante/est-robot · andreyviolante.github.io/est-robot

- Reverse-engineered the USB (HID) protocol and binary format of an educational robot with no public documentation.
- Wrote a compiler from Python to the robot bytecode, tested byte by byte against the official IDE: 13 out of 13 programs identical.
- Designed a Portuguese-language API for classes, with explained errors, an in-browser editor (Python or blocks) and a simulator.
- Open source (MIT), with automated tests on GitHub Actions, a teacher guide and a Zenodo DOI (10.5281/zenodo.23692767).

Hi, Kiara · React Native, Expo, TypeScript, Vercel, Postgres, LLM
hi-kiara.vercel.app

- Virtual assistant for 6th to 9th graders to practice English by voice and text, using a Socratic method and safety rules for minors.
- Serverless backend: the model API key stays on the server, with a fallback model when the main one hits its rate limit.
- Teacher dashboard with classes stored in Postgres; student conversations are never stored.
- Unit and browser tests, with automatic deployment only after they pass. Presented at the 6th Maricá Pedagogical Fair (2026).

Other projects

- **Finance-Ai:** Android app (React Native and Java) that logs expenses by reading bank notifications, with data kept on the device.
- **RegionWatcher:** Android app (Kotlin and Jetpack Compose) that alerts when an area of the screen changes, comparing perceptual hashes.
- **Moodle-Assistente:** Python desktop app that sorts college assignments by urgency and opens an AI tutor.
- **Estudos-Backend:** college exercises with FastAPI, PostgreSQL, MongoDB, RabbitMQ, Kafka and Docker.

EXPERIENCE

Robotics Teacher · City of Maricá (Prefeitura Municipal de Maricá) · Aug 2023 – present

- Robotics and programming classes for elementary and middle school students.
- Coaching a team for the Brazilian Robotics Olympiad (OBR).
- Building the tools used in class, such as EST-LAB.

EDUCATION

B.S. in Software Engineering · Universidade de Vassouras · 6th semester, in progress

SKILLS

Backend	Python, FastAPI, Django, SQLAlchemy, pytest, REST APIs
Data and messaging	PostgreSQL, MongoDB, MySQL, RabbitMQ, Kafka
Mobile and front end	React Native, Expo, TypeScript, Kotlin, Jetpack Compose
Tools	Docker, Git, GitHub Actions, Linux, Vercel, Arduino
Languages	Portuguese (native), English (advanced)